Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 20: Approved Solutions

Bijections, Inverses, and Composition

Covers Chartrand & Zhang, Section 5.4

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first.

Part I: How to Practice This Material

Bijective means invertible

The equivalence to keep in mind

$f : A \to B$ has an inverse **if and only if** f is bijective.
Onto guarantees $f^{-1}(b)$ is defined for every b. **One-to-one** guarantees it is defined *uniquely*. Those are exactly the two conditions for f^{-1} to be a function.

Finding an inverse

Write $y = f(x)$, solve for x in terms of y, and that expression is $f^{-1}(y)$. If you cannot solve uniquely, the function was not bijective.

Three mistakes to expect

1. **Writing f^{-1} for a non-bijective function.** It does not exist.
2. **Confusing f^{-1} with $1/f$.** Different things entirely.
3. **Assuming $g \circ f = f \circ g$.** Composition is not commutative.

Part II: Practice Problems

Section A — Bijections and inverses

1. For each function, decide whether it is bijective. If it is, give f^{-1}; if not, say which property fails and give a witness.

(a) $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 6 - 2x$.

(b) $g : \mathbb{Z} \to \mathbb{Z}$, $g(n) = n^2$.

(c) $h : \mathbb{R} \to \mathbb{R}$, $h(x) = x^3 + 1$.

Approved solution

(a) Bijective. One-to-one: $6-2a = 6-2b$ gives $-2a = -2b$, so $a = b$. Onto: given b, take $a = \dfrac{6-b}{2}$; then $f(a) = 6 - (6 - b) = b$.

$$f^{-1}(y) = \frac{6-y}{2}$$

(b) Not bijective. Not one-to-one: $g(2) = g(-2) = 4$. Not onto either: -1 has no preimage, and neither does 2. One failing property is enough.

(c) Bijective. One-to-one: $a^3+1 = b^3+1$ gives $a^3 = b^3$, and taking cube roots gives $a = b$ (cube root is defined and single-valued on all of $\mathbb{R}$). Onto: given b, take $a = \sqrt[3]{b-1}$; then $h(a) = (b-1)+1 = b$.

$$h^{-1}(y) = \sqrt[3]{y-1}$$

Note the contrast between (b) and (c). Squaring loses sign information and cubing does not — which is why x^3 is a bijection on $\mathbb{R}$ and x^2 is not.

2. Let $f : \mathbb{R} \to \mathbb{R}$ be $f(x) = 4x - 9$. Verify directly that $f^{-1}(f(x)) = x$ and $f(f^{-1}(y)) = y$.

Approved solution

Solving $y = 4x - 9$ gives $x = \dfrac{y+9}{4}$, so $f^{-1}(y) = \dfrac{y+9}{4}$.

$$f^{-1}(f(x)) = \frac{(4x-9)+9}{4} = \frac{4x}{4} = x \qquad f(f^{-1}(y)) = 4\left(\frac{y+9}{4}\right) - 9 = (y+9) - 9 = y$$

Both identities hold, as the theorem requires. **Checking both directions is the point** — a function can undo another on one side without being a two-sided inverse, and for bijections both must hold.

Section B — Composition

3. Let $f, g : \mathbb{R} \to \mathbb{R}$ with $f(x) = 2x - 1$ and $g(x) = x + 3$.

(a) Find $g \circ f$ and $f \circ g$.

(b) Is $g \circ f = f \circ g$? If not, find where they agree.

(c) Find $(g \circ f)^{-1}$ and compare it with $f^{-1} \circ g^{-1}$.

Approved solution

(a) $(g \circ f)(x) = (2x - 1) + 3 = 2x + 2$. $\quad (f \circ g)(x) = 2(x + 3) - 1 = 2x + 5$.

(b) No — $2x + 2 = 2x + 5$ has no solution, so they agree *nowhere*. The two compositions are parallel lines.

(c) Solving $y = 2x + 2$ gives $x = \dfrac{y-2}{2}$, so $(g \circ f)^{-1}(y) = \dfrac{y-2}{2}$.
Now $f^{-1}(y) = \dfrac{y+1}{2}$ and $g^{-1}(y) = y - 3$, so

$$(f^{-1} \circ g^{-1})(y) = f^{-1}(y-3) = \frac{(y-3)+1}{2} = \frac{y-2}{2}.$$

They are equal: $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Note the order reverses. To undo "f then g" you undo g first, then f — the same reason you remove your shoes before your socks. This is a general theorem, and part (c) is a worked instance of it.

4. Let $f : A \to B$ and $g : B \to C$ both be one-to-one. Prove that $g \circ f$ is one-to-one.

Approved solution

Assume $(g \circ f)(a) = (g \circ f)(b)$ for $a, b \in A$. By definition of composition this says $g(f(a)) = g(f(b))$.
Since g is one-to-one, applying its defining property to the elements $f(a)$ and $f(b)$ of B gives $f(a) = f(b)$.
Since f is one-to-one, this gives $a = b$.
Therefore $g \circ f$ is one-to-one. ■

The structure is worth naming: peel from the outside in. The outermost function is undone first, then the inner one. Three lines, each one application of the working form of the definition.

Section C — The induced equivalence relation

5. Let $f : \{1,2,3,4,5,6\} \to \{a,b,c\}$ be $f = \{(1,a),(2,b),(3,a),(4,c),(5,b),(6,a)\}$. Define $x \sim y$ if $f(x) = f(y)$.

(a) List the equivalence classes of $\sim$.

(b) Is f one-to-one? How do the classes show it?

(c) Is f onto? How do the classes show it?

Approved solution

(a) $[1] = \{1,3,6\}$, $[2] = \{2,5\}$, $[4] = \{4\}$. Three classes.
(b) Not one-to-one. A class with more than one element means two distinct inputs share an image — here 1, 3, and 6 all map to a. **f is one-to-one exactly when every class is a singleton.**
(c) Onto. There are three classes and three elements in the codomain, and each class maps to a different one. In general f is onto exactly when the number of classes equals $|B|$ — since each class corresponds to one attained value.

The classes are the preimages. $[1] = f^{-1}(\{a\})$, and so on. This is the correspondence from Lecture 20, and it makes both properties visible at a glance: singleton classes mean one-to-one, and a class for every codomain element means onto.

Section D — Writing

6. In five to eight complete sentences, explain why a function must be both one-to-one and onto to have an inverse. Address each property separately, and connect your answer to what it takes for a relation to be a function at all.

Approved solution

Model answer. To define $f^{-1}(b)$ we must send each b in the codomain back to the element of the domain that maps to it, so f^{-1} is a relation from B to A, and for it to be a *function* every element of B must have exactly one image. That phrase splits into two requirements, which is why two properties are needed. Onto supplies existence: if some b were the image of nothing, then $f^{-1}(b)$ would be undefined and the first failure mode from Lecture 18 would occur. One-to-one supplies uniqueness: if two distinct elements a and a' both mapped to b, then $f^{-1}(b)$ would have two candidate values, which is the second failure mode. So bijectivity is not an extra condition layered on top — it is exactly the translation of "f^{-1} is a function" back into a statement about f.

Full credit requires matching each property to one of the two failure modes for being a function. An answer that says only "you need both" without explaining which does what earns about half.

Section E — Challenge (optional)

7. Prove that if $f : A \to B$ and $g : B \to C$ are both bijective, then $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Approved solution

Both $g \circ f$ and $f^{-1} \circ g^{-1}$ are functions from C to A and A to C respectively, and both compositions are bijective, so both inverses exist.
Let $c \in C$. We show $(f^{-1} \circ g^{-1})(c)$ is the element that $g \circ f$ sends to c. Write $b = g^{-1}(c)$ and $a = f^{-1}(b)$, so $(f^{-1} \circ g^{-1})(c) = a$. Then

$$(g \circ f)(a) = g\big(f(a)\big) = g\big(f(f^{-1}(b))\big) = g(b) = g\big(g^{-1}(c)\big) = c.$$

So $g \circ f$ maps a to c, which means $(g \circ f)^{-1}(c) = a$. Since c was arbitrary, the two functions agree everywhere and are equal. ■

Why the order reverses. $g \circ f$ does f first and g second; undoing it must undo the *last* step first. Socks then shoes, so shoes off then socks off.
Problem 3(c) was this theorem in a specific case. Doing the example first and the proof second is a reasonable order — the example tells you what to prove.

Instructor notes

- **Problem 5 is the loop-closer.** Equivalence classes make one-to-one and onto visible at a glance: singleton classes mean injective, one class per codomain element means surjective. Worth board time — it unifies Weeks 6 and 7.
- **Problems 3(c) and 7 are the same result** at two levels. Assign 3(c) to everyone; 7 to whoever wants the general argument.
- **Problem 6** is the writing anchor and the best statement of why bijectivity is not an arbitrary

hypothesis. It maps each property onto one of Lecture 18's two failure modes.

- **Problem 1(b)** has two failing properties; students need name only one. Watch for over-answering, which wastes time on the exam.
- **Problem 4** teaches "peel from the outside in," which is the shape of most composition proofs.
- Reserved for Homework 8: Section 5.4 even-numbered exercises.